Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/16/2025
PI Name:	Boerma, Marjan Ph.D.		
Project Title:	Gamma-Tocotrienol as a Countermeasure against High-Energy Charged Particle-Induced Carcinogenesis, Cardiovascular Disease, and Central Nervous System Effects		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Little Rock	**State:**	AR
Zip Code:	72205-7101	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/31/2019	**End Date:**	10/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2025 per NSSC information (Ed., 9/17/24) NOTE: End date changed to 10/31/2024 per M. Kirby/JSC and NSSC information (Ed., 8/14/23) NOTE: End date changed to 10/31/2023 per V. Lehman/JSC (Ed., 6/20/23) NOTE: End date changed to 10/31/2022 per NSSC information (Ed., 5/17/21) NOTE: End date changed to 3/31/2022 per NSSC information (Ed., 11/4/20)		
Key Personnel Changes/Previous PI:	Ed. note - PI addition to Nov 2021 report: We are currently in a phase of the grant in which Drs. Pathak and Landes are no longer Co-Investigators. At this point in the project, only Dr. Weil is a Co-Investigator (CoI). No changes in Principal Investigator (PI) or other key personnel.		
COI Name (Institution):	Weil, Michael Ph.D. (Colorado State University)		
Grant/Contract No.:	80NSSC19K0437		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Recent evidence shows that radiation encountered during deep space travel is associated with increased risks of cancer. Administration of a dietary radiation countermeasure before and/or during the mission is an attractive option to reduce the carcinogenesis risk. Gamma-tocotrienol is one of the strongest radiation protectors of all natural compounds tested so far and it has shown cancer prevention properties in human subjects and animal models. It is safe, non-toxic and well tolerated, exhibits no interactions with other medications, and requires no special storage conditions. Altogether, gamma-tocotrienol has high potential as a radiation countermeasure during space travel. In this project, we develop a mouse model of radiation carcinogenesis that may be used to test the protective properties of gamma-tocotrienol and other countermeasures. For this purpose, we are searching for a genetically modified mouse model that shows a low spontaneous cancer rate, but increased tumor incidence in response to low-dose radiation. P53deltaP mice carry a mutation in the P53 gene that makes them more susceptible to carcinogenesis. We obtained P53deltaP mice and created a breeding colony. However, at around 100 days of age, these mice started developing a range of hematologic and solid tumors, and this unexpectedly high rate of tumor formation in untreated animals made the mice unsuitable for studying long-term carcinogenesis from low doses of ionizing radiation. Therefore, a major goal of the project is to cross P53deltaP mice onto different genetic backgrounds that we expect will lower the number of spontaneous tumors. We will monitor survival time and tumor formation with and without low-dose radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There is concern about increased carcinogenesis risk after chronic exposures to low-dose ionizing radiation, such as from medical treatments, occupational low-dose exposures, and radiological accidents. This project aims to develop a mouse model of low-dose radiation-induced carcinogenesis that may be used to test gamma-tocotrienol and other countermeasures. This information will not only contribute to reducing the risk of radiation exposure during deep-space travel, but also the risks of carcinogenesis from exposure to low-dose rate radiation on Earth.
Task Progress:	In prior project years, P53deltaP mice on an FVB/Jax genetic background were obtained and used to create a breeding colony. However, at around 100 days of age, these mice started developing a range of hematologic and solid tumors, including lung tumors, mammary tumors, and subcutaneous tumors. The unexpectedly high rate of tumor formation in untreated animals make P53deltaP mice on an FVB/Jax genetic background unsuitable for studying long-term carcinogenesis from low doses of ionizing radiation. Therefore, we crossed P53deltaP mice on a C57BL/6J genetic background, and we crossed P53deltaP mice on a mixed C57BL/6J and 129S4/SvJae background. We expect that P53deltaP mice on one or both genetic backgrounds will show low numbers of spontaneous tumors but a wide range of tumor types after low-dose radiation exposure. First, P53deltaP mice on both genetic backgrounds were housed for aging and long-term follow up. The average survival time of P53deltaP mice on a C57BL/6J background and the mixed background was 470 days and 489 days, respectively. These results suggest that the background tumor formation in P53deltaP mice on both genetic backgrounds is lower compared to the original line of these mice. Next, we exposed P53deltaP mice of both genetic backgrounds to low dose (4 Gy) gamma-rays and observed a significant decrease in survival and increased rate of tumor formation, suggesting that both mouse lines create good models to study carcinogenesis from low-dose radiation.
Bibliography Type:	Description: (Last Updated: 01/08/2026)
Conference Materials (Downloadable)	Boerma M, Sridharan V, Landes RD, Weil MM. "Gamma-tocotrienol as a countermeasure against high-energy charged particle-induced carcinogenesis." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 GT3 poster HRP 2025 final.pdf